MACHINE LEARNING LABORATORY

Laboratory Experiment – 06 (13th Oct, 2026)

Implementation and Comparative Performance Analysis of Traditional Statistical and Conventional Machine Learning Regression Models for FBG-Based Tilt Angle Prediction

Duration: 3 Hours
Total Marks: 100
Dataset: FBG Biaxial Tilt Sensor Dataset
Tools: Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn

1. Aim

To implement and evaluate traditional statistical regression and conventional machine learning regression algorithms for predicting axial and transverse tilt angles using Fiber Bragg Grating (FBG) sensor measurements.

Students will compare the prediction accuracy, generalization capability, and computational efficiency of different regression techniques.

2. Dataset Description

The provided Excel dataset contains three worksheets:

Dataset link:`https://shorturl.at/xKTin`

Worksheet	Description
Theoratical calculations	Theoretical sensor-response calculations
Experiment axial	Experimental measurements for axial tilt
Experiment transverse	Experimental measurements for transverse tilt

The axial and transverse experimental worksheets contain 37 observations each. Each experimental worksheet contains **37 observations** and **seven principal data columns**, with the first row representing a measurement rather than a header.

Table 1: Description of Experimental Dataset Columns

Column	Description
A	Tilt angle, θ (degrees)
B	FBG1 wavelength or spectral-peak measurement
C	FBG2 wavelength or spectral-peak measurement
D	Change in FBG1 measurement
E	Change in FBG2 measurement
F	Differential wavelength change
G	Average wavelength change

The principal data columns contain tilt angle, FBG sensor measurements, wavelength changes, differential wavelength changes, and average wavelength changes.

The regression problem is formulated as:

$$X = [\Delta\lambda_1, \Delta\lambda_2] \tag{1}$$

$$Y = \theta \tag{2}$$

where $\Delta\lambda_1$ and $\Delta\lambda_2$ represent the wavelength changes, and θ represents the axial or transverse tilt angle in degrees.

The axial and transverse datasets must be analyzed separately.

3. Task and Time Allocation

Task	Description	Time	Marks
1	Dataset exploration and statistical analysis	20 min	10
2	Data visualization and preprocessing	20 min	10
3	Feature engineering and feature selection	20 min	10
4	Traditional statistical regression models	30 min	20
5	Conventional ML regression models	40 min	25
6	Performance evaluation and comparison	35 min	20
7	Result interpretation and conclusions	15 min	5
Total		**180 min**	**100**

4. Laboratory Tasks

Task 1: Dataset Exploration and Statistical Analysis (20 Minutes | 10 Marks)

Objective: Understand the statistical characteristics of the experimental FBG sensor measurements.

1. Load the axial and transverse experimental worksheets using Pandas.
2. Identify the total number of observations, features, missing values, and duplicate records.
3. Calculate the mean, median, standard deviation, variance, minimum, and maximum for each numerical feature.
4. Examine the distribution of axial and transverse tilt angles.
5. Identify repeated tilt angles and compare their corresponding wavelength measurements.

Required Outputs: Dataset summary, descriptive statistics, and feature distribution plots.

Task 2: Data Visualization and Preprocessing (20 Minutes | 10 Marks)

1. Plot FBG1 and FBG2 wavelength responses against axial tilt angle.
2. Repeat the plots for transverse tilt measurements.
3. Generate a correlation heatmap for both datasets.
4. Identify possible outliers using box plots.

5. Apply StandardScaler and MinMaxScaler separately to the input features.
6. Compare feature distributions before and after normalization.

Required Outputs: Scatter plots, correlation heatmaps, box plots, and normalized feature distributions.

Note: All data scaling operations used for model evaluation must be fitted only on the training data.

Task 3: Feature Engineering and Feature Selection (20 Minutes | 10 Marks)

1. Compute the differential wavelength change:

$$\Delta\lambda_{\text{diff}} = \Delta\lambda_1 - \Delta\lambda_2 \tag{3}$$

2. Compute the average wavelength change:

$$\Delta\lambda_{\text{avg}} = \frac{\Delta\lambda_1 + \Delta\lambda_2}{2} \tag{4}$$

3. Prepare the following feature configurations:
   - F1: Individual FBG wavelength changes
   - F2: Differential wavelength change
   - F3: Average wavelength change
   - F4: Differential and average wavelength changes
4. Calculate Pearson correlation coefficients between the engineered features and tilt angle.
5. Identify the most informative feature configuration for axial and transverse tilt prediction.

Required Outputs: Feature comparison table and correlation analysis.

Task 4: Traditional Statistical Regression Models (30 Minutes | 20 Marks)

Implement the following regression models:

1. Simple Linear Regression (SLR)
2. Multiple Linear Regression (MLR)
3. Polynomial Regression (Degrees 2 and 3)
4. Ridge Regression
5. Lasso Regression
6. Elastic Net Regression

For each model:

(a) Train the model using the selected input features.

(b) Predict axial and transverse tilt angles separately.

(c) Calculate MAE, MSE, RMSE, and R^2.

(d) Analyze the effects of polynomial degree and regularization.

(e) Generate actual-versus-predicted angle plots.

Required Outputs: Regression performance table, prediction plots, and comparison of statistical models.

Task 5: Conventional Machine Learning Regression Models (40 Minutes | 25 Marks)

Implement the following regression algorithms:

Model	Hyperparameters
Support Vector Regression (SVR)	Linear and RBF kernels
Decision Tree Regression (DTR)	Maximum depth: 2, 3, 5
Random Forest Regression (RFR)	Number of trees: 50, 100
K-Nearest Neighbors (KNN)	Number of neighbors: 3, 5, 7
Gradient Boosting Regression (GBR)	Learning rate: 0.01, 0.1
Gaussian Process Regression (GPR)	RBF and Matérn kernels

Perform the following:

1. Implement and train all six regression models using the experimental datasets.
2. Investigate the effect of different hyperparameter values.
3. Compare prediction accuracy for axial and transverse tilt estimation.
4. Analyze prediction errors for individual tilt angles.
5. Identify the best-performing conventional machine learning regression model.

Required Outputs: Model performance table, hyperparameter comparison, and residual plots.

Task 6: Performance Evaluation and Comparative Analysis (35 Minutes | 20 Marks)

Evaluate the models using five-fold grouped cross-validation. Measurements corresponding to the same tilt angle must remain in the same fold.

Hyperparameter selection must be performed within training folds to prevent data leakage.

Calculate the following performance metrics:

Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{5}$$

Mean Squared Error (MSE):

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 \tag{6}$$

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{8}$$

Prepare separate comparison tables for axial and transverse tilt prediction.

Model	MAE	MSE	RMSE	R^2	Training Time (s)
Linear Regression					
Polynomial Regression					
Ridge Regression					
Lasso Regression					
! Elastic Net					
SVR					
Decision Tree					
Random Forest					
KNN					
Gradient Boosting					
GPR					

Generate the following graphs:

1. Model-wise MAE comparison
2. Model-wise RMSE comparison
3. Model-wise R^2 comparison
4. Actual versus predicted angles for the best statistical regression model
5. Actual versus predicted angles for the best conventional ML regression model
6. Residual plots for the two selected models
7. Training time comparison across regression models

Required Outputs: Performance comparison tables, seven graphs, and a brief interpretation of the results.

Task 7: Result Interpretation and Conclusions (15 Minutes | 5 Marks)

Answer the following questions based on the experimental findings:

1. Which traditional statistical regression model provides the best prediction accuracy?
2. Which conventional ML regression model achieves the lowest RMSE?
3. Do nonlinear regression algorithms outperform linear regression algorithms?
4. How does feature selection influence regression performance?
5. Does increasing model complexity always improve prediction accuracy?
6. Why might prediction errors differ between axial and transverse tilt measurements?
7. Which regression model would you recommend for real-time FBG-based tilt monitoring? Justify your answer.

5. Any one Advanced Task to be performed

Task A: Noise Robustness

Introduce Gaussian noise into wavelength measurements and investigate its effect on prediction accuracy. Apply noise only to the evaluation inputs.

Task B: Sensor Hysteresis

Compare wavelength measurements recorded at identical tilt angles during different measurement sequences. Analyze possible hysteresis behavior.

Task C: Signed and Absolute Tilt Prediction

For the transverse dataset, compare regression performance when predicting the signed tilt angle θ and absolute tilt magnitude $|\theta|$.

Discuss whether the available sensor measurements can reliably distinguish positive and negative tilt.

6. Submission Instructions

Students must submit a single Report(only pdf) of max 6 pages containing:

1. Methodology, diagram with any one model end to end process, Algorithm with any one model, Result in tabular form with all observation with different models, and short Discussion.
2. All required graphs and visualizations.
3. code link it should be public access.
4. Regression performance comparison tables.
5. Hyperparameter configurations and results.
6. Advance task observation.

Upload file link:`https://forms.gle/RCiyByKQtEbeoNVB7`
Important Instructions:

- Use the provided dataset for all experiments.
- Use a consistent validation methodology for comparing regression models.
- Prevent data leakage during preprocessing, feature selection, and hyperparameter tuning.
- Specify random seeds wherever applicable.
- Do not artificially increase the number of experimental observations.
- Do not merge theoretical and experimental samples for model evaluation.
- All reported results must be generated through actual program execution.

7. Expected Learning Outcomes

After completing this experiment, students will be able to:

1. Analyze experimental sensor datasets.
2. Perform preprocessing and feature engineering.

3. Implement statistical regression algorithms.
4. Implement conventional machine learning regression algorithms.
5. Evaluate models using appropriate metrics.
6. Investigate overfitting and generalization.
7. Compare predictive accuracy and computational efficiency.
8. Select suitable models for sensor calibration and tilt-angle estimation.